

Integers, Place Value and Arithmetic

Module 2 · In-class Sheet

Goal: Answer integer and place-value questions fast, with zero errors.

Worked Example 1 — Restructure before you calculate

Example:

Problem. What is the value of $36 \times 7 + 64 \times 7$?

Scan first (10 seconds). Both products contain $\times 7$. 36 and 64 make 100.

Restructure. $36 \times 7 + 64 \times 7 = (36 + 64) \times 7 = 100 \times 7$.

Calculate. $100 \times 7 = 700$.

Check. 36×7 is about 250 and 64×7 is about 450; $250 + 450 = 700$. ✓

Lesson. A repeated factor or a pair that makes a round number is a signal to regroup.

Worked Example 2 — Two methods, one answer: which is faster?

Example:

Problem. What is $(2 + 4 + 6 + \dots + 50) - (1 + 3 + 5 + \dots + 49)$?

Method A — calculate each sum. Evens: $2 \times (1 + 2 + \dots + 25) = 2 \times 325 = 650$.

Odds: $1 + 3 + \dots + 49 = 25 \times 25 = 625$. Difference: $650 - 625 = 25$.

Method B — pair the terms. $(2 - 1) + (4 - 3) + \dots + (50 - 49)$. There are 25 pairs and each pair is 1, so the answer is 25.

Compare. Method A needs two sum formulas and a subtraction. Method B needs one count. Both give 25; Method B is the contest method.

Lesson. When two long lists line up term by term, subtract in pairs.

Guided practice

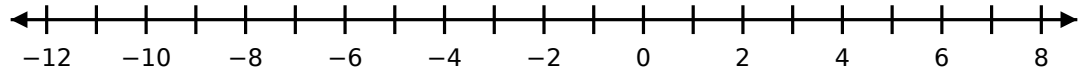
Lesson A · 20 min. Do these together; say the scaffold step aloud first.

- In 7 350 912, which digit is in the ten-thousands place? *Scaffold: write the digits into the chart, starting from the ones (O) column.*

M	HTh	TTh	Th	H	T	O

2. Which of these numbers is farthest from 0? -8.2 , 7.9 , -0.85 , 8.1 , -8.02 *Scaffold: write the distance from 0 of each number first.*

3. At 5 a.m. the temperature was -11°C . By 2 p.m. it was 6°C . By how many degrees did the temperature rise? *Scaffold: draw a jump to 0, then a jump to 6.*



4. A runner's two laps took 3 min 48 s and 4 min 25 s. What was the total time? *Scaffold: add the seconds first; trade 60 s for 1 min.*

5. Calculate $125 \times 13 \times 8$. *Scaffold: which two factors make a round number?*

6. Calculate $19 \times 46 + 81 \times 46$. *Scaffold: what do both products share?*

Independent practice

Lesson A (paired, Q7-10) and Lesson B (independent, Q11-14). Target: 30 seconds per question, zero errors.

7. When the numbers 0.305, 0.35, 0.3, 0.053 and 0.53 are listed from smallest to largest, which number is in the middle?
- (A) 0.305 (B) 0.35 (C) 0.3 (D) 0.053 (E) 0.53

8. What is the sum of the digits of the number 4 070 809?
(A) 19 (B) 24 (C) 27 (D) 28 (E) 29

9. A bakery sells muffins for \$2.35 each. What is the cost of 12 muffins?
(A) \$23.50 (B) \$25.85 (C) \$28.20 (D) \$28.70 (E) \$30.55

10. Two boards are each 360 cm long. One is cut completely into 12 cm pieces and the other into 20 cm pieces. How many more 12 cm pieces are there than 20 cm pieces?
(A) 8 (B) 12 (C) 18 (D) 30 (E) 48

11. Which number is equal to $4 \times 10^4 + 3 \times 10^2 + 5 \times 10$?
(A) 4 350 (B) 40 350 (C) 43 050 (D) 400 350 (E) 403 500

12. What is the value of $12 - (-7) + (-15)$?
(A) -10 (B) -4 (C) 4 (D) 20 (E) 34

13. On a standard die the numbers 1 to 6 appear once each, so the six faces add to 21. A die sits on a table. The five faces that are not touching the table add to 17. What number is on the face touching the table?
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

14. A movie starts at 7:48 p.m. and runs for 2 hours 37 minutes. At what time does it end?
(A) 9:25 p.m. (B) 9:45 p.m. (C) 10:05 p.m. (D) 10:15 p.m. (E) 10:25 p.m.

★ **Challenge**

Stretch — Part C level. Record every case you try in the space.

15. The number $10^{30} - 47$ is written out in full. What is the sum of its digits? *Use the space to build a table of small cases: $10^2 - 47$, $10^3 - 47$, $10^4 - 47$.*
(A) 252 (B) 259 (C) 260 (D) 262 (E) 270

16. How many digits does the number $5^{20} \times 2^{18}$ have when it is written out in full?
(A) 19 (B) 20 (C) 21 (D) 38 (E) 40

Strategy triggers:

When you see **two products sharing a number**, reach for **factoring it out**: $a \times c + b \times c = (a + b) \times c$.

When you see **numbers close to a round number**, reach for **compensation**: write each as round $\pm$ small.

When you see **25, 125 or 5** in a product, reach for **their partners 4, 8 or 2**.

When you see **a change that crosses 0**, reach for **two jumps: to 0, then onward**.

When you see **10^n minus something**, reach for **small cases $n = 2, 3, 4$** and count the 9s.

Exit ticket

Lesson B · last 10 min. On your own.

17. Calculate $250 \times 17 \times 4$.

18. Find $999 + 998 + 1002 + 1001$ and explain in one or two sentences why your method is fast.

Integers, Place Value and Arithmetic

Module 2 · In-class Sheet · Teacher Version

Original Concept Mastery items. Not reproduced from CEMC or any third party.

Goal: Answer integer and place-value questions fast, with zero errors.

Worked Example 1 — Restructure before you calculate

Example:

Problem. What is the value of $36 \times 7 + 64 \times 7$?

Scan first (10 seconds). Both products contain $\times 7$. 36 and 64 make 100.

Restructure. $36 \times 7 + 64 \times 7 = (36 + 64) \times 7 = 100 \times 7$.

Calculate. $100 \times 7 = 700$.

Check. 36×7 is about 250 and 64×7 is about 450; $250 + 450 = 700$. ✓

Lesson. A repeated factor or a pair that makes a round number is a signal to regroup.

Worked Example 2 — Two methods, one answer: which is faster?

Example:

Problem. What is $(2 + 4 + 6 + \dots + 50) - (1 + 3 + 5 + \dots + 49)$?

Method A — calculate each sum. Evens: $2 \times (1 + 2 + \dots + 25) = 2 \times 325 = 650$.

Odds: $1 + 3 + \dots + 49 = 25 \times 25 = 625$. Difference: $650 - 625 = 25$.

Method B — pair the terms. $(2 - 1) + (4 - 3) + \dots + (50 - 49)$. There are 25 pairs and each pair is 1, so the answer is 25.

Compare. Method A needs two sum formulas and a subtraction. Method B needs one count. Both give 25; Method B is the contest method.

Lesson. When two long lists line up term by term, subtract in pairs.

Guided practice

Lesson A · 20 min. Do these together; say the scaffold step aloud first.

Timing: 20 min (Lesson A). Cut if long: Q5 (Q6 tests the same structure).

1. In 7 350 912, which digit is in the ten-thousands place? *Scaffold: write the digits into the chart, starting from the ones (O) column.*

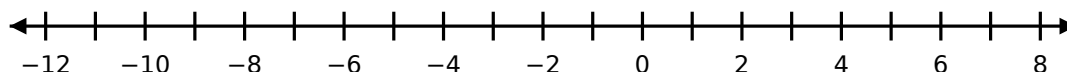
M	HTh	TTh	Th	H	T	O

Answer 5 · Likely wrong: 3 (one place too far: hundred-thousands) or 9 (counted five places from the left) · Target 20 s · place value, level 1

2. Which of these numbers is farthest from 0? -8.2 , 7.9 , -0.85 , 8.1 , -8.02 *Scaffold: write the distance from 0 of each number first.*

Answer -8.2 · Likely wrong: -0.85 or 8.1 — chose the smallest-looking or the largest positive number · Target 25 s · integers / decimals, level 1

3. At 5 a.m. the temperature was -11°C . By 2 p.m. it was 6°C . By how many degrees did the temperature rise? *Scaffold: draw a jump to 0, then a jump to 6.*



Answer 17 · Likely wrong: 5 — computed $11 - 6$ (dropped the sign) · Target 25 s · integers, level 1

4. A runner's two laps took 3 min 48 s and 4 min 25 s. What was the total time? *Scaffold: add the seconds first; trade 60 s for 1 min.*

Answer 8 min 13 s · Likely wrong: 7 min 73 s or 8 min 73 s — no regrouping, or regrouped by 100 · Target 30 s · time, level 1

5. Calculate $125 \times 13 \times 8$. *Scaffold: which two factors make a round number?*

Answer 13 000 · Likely wrong: $1\,625 \times 8$ done by hand and slipped (e.g. 12 000) · Target 20 s · efficient arithmetic, level 2

6. Calculate $19 \times 46 + 81 \times 46$. *Scaffold: what do both products share?*

Answer 4 600 · Likely wrong: 4 646 or an arithmetic slip from computing $874 + 3726$ by hand · Target 20 s · efficient arithmetic, level 2

Independent practice

Lesson A (paired, Q7–10) and Lesson B (independent, Q11–14). Target: 30 seconds per question, zero errors.

Timing: Q7–10 paired 12 min (Lesson A); Q11–14 independent 8 min (Lesson B). Cut if long: Q10 and Q13 (move to the extension session).

7. When the numbers 0.305, 0.35, 0.3, 0.053 and 0.53 are listed from smallest to largest, which number is in the middle?

(A) 0.305 (B) 0.35 (C) 0.3 (D) 0.053 (E) 0.53

Answer (A) 0.305 · Likely wrong: (B) 0.35 — believes more digits means larger ($0.305 > 0.35$) · Target 30 s · decimals, level 1

8. What is the sum of the digits of the number 4 070 809?

(A) 19 (B) 24 (C) 27 (D) 28 (E) 29

Answer (D) 28 · Likely wrong: (A) 19 — dropped the final 9; (B) 24 — dropped the leading 4 · Target 20 s · digit sums, level 1

9. A bakery sells muffins for \$2.35 each. What is the cost of 12 muffins?

(A) \$23.50 (B) \$25.85 (C) \$28.20 (D) \$28.70 (E) \$30.55

Answer (C) \$28.20 · Likely wrong: (B) \$25.85 — $10 \times 2.35 + 2.35$ (added one muffin, not two) · Target 30 s · money, level 1

10. Two boards are each 360 cm long. One is cut completely into 12 cm pieces and the other into 20 cm pieces. How many more 12 cm pieces are there than 20 cm pieces?

(A) 8 (B) 12 (C) 18 (D) 30 (E) 48

Answer (B) 12 · Likely wrong: (A) 8 — subtracted the lengths $20 - 12$ · Target 30 s · division, level 2

11. Which number is equal to $4 \times 10^4 + 3 \times 10^2 + 5 \times 10$?

(A) 4 350 (B) 40 350 (C) 43 050 (D) 400 350 (E) 403 500

Answer (B) 40 350 · Likely wrong: (C) 43 050 — shifted the 3 one place left · Target 25 s · powers of ten, level 1

12. What is the value of $12 - (-7) + (-15)$?

(A) -10 (B) -4 (C) 4 (D) 20 (E) 34

Answer (C) 4 · Likely wrong: (A) -10 — treated $-(-7)$ as -7 · Target 25 s · integers, level 1

13. On a standard die the numbers 1 to 6 appear once each, so the six faces add to 21. A die sits on a table. The five faces that are not touching the table add to 17. What number is on the face touching the table?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Answer (D) 4 · Likely wrong: (C) 3 — gave the top face ($7 - 4$) instead of the bottom face · Target 25 s · complements, level 1

14. A movie starts at 7:48 p.m. and runs for 2 hours 37 minutes. At what time does it end?

(A) 9:25 p.m. (B) 9:45 p.m. (C) 10:05 p.m. (D) 10:15 p.m. (E) 10:25 p.m.

Answer (E) 10:25 p.m. · Likely wrong: (A) 9:25 p.m. — did not carry the extra hour from 85 minutes · Target 30 s · time, level 2

★ Challenge

Stretch — Part C level. Record every case you try in the space.

Timing: 12 min (Lesson B, paired). Cut if long: Q16 — keep Q15, it is the strand's one Part C structure.

15. The number $10^{30} - 47$ is written out in full. What is the sum of its digits? *Use the space to build a table of small cases: $10^2 - 47$, $10^3 - 47$, $10^4 - 47$.*

(A) 252 (B) 259 (C) 260 (D) 262 (E) 270

Answer (C) 260 · Likely wrong: (E) 270 — assumed 30 nines; (A) 252 — 28 nines but dropped the final 53 · Target 150 s · $10^n - k$, level 4

16. How many digits does the number $5^{20} \times 2^{18}$ have when it is written out in full?

- (A) 19 (B) 20 (C) 21 (D) 38 (E) 40

Answer (B) 20 · Likely wrong: (D) 38 — added the exponents; (C) 21 — used 10^{20} instead of 25×10^{18} · Target 120 s · powers of ten, level 4

Strategy triggers:

When you see **two products sharing a number**, reach for **factoring it out**: $a \times c + b \times c = (a + b) \times c$.

When you see **numbers close to a round number**, reach for **compensation**: write each as round $\pm$ small.

When you see **25, 125 or 5** in a product, reach for **their partners 4, 8 or 2**.

When you see **a change that crosses 0**, reach for **two jumps: to 0, then onward**.

When you see **10^n minus something**, reach for **small cases $n = 2, 3, 4$** and count the 9s.

Exit ticket

Lesson B · last 10 min. On your own.

Timing: 10 min (Lesson B). Never cut.

17. Calculate $250 \times 17 \times 4$.

Answer 17 000 · Likely wrong: 4 250×4 slipped (e.g. 16 000) · Target 20 s · efficient arithmetic, level 1

18. Find $999 + 998 + 1002 + 1001$ and explain in one or two sentences why your method is fast.

Answer 4 000 · Likely wrong: An addition slip from column-adding four 4-digit numbers · Target 40 s · efficient arithmetic, level 2